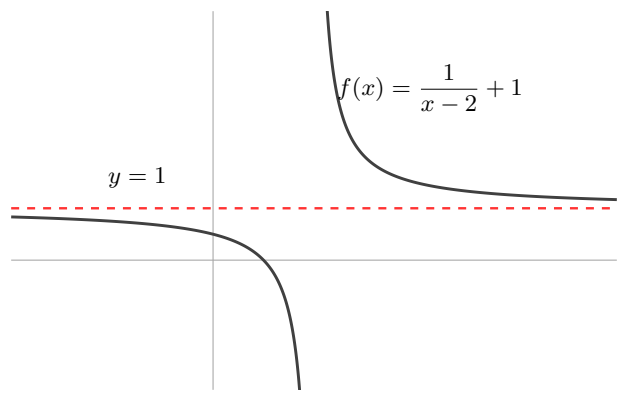




A graph of the function $f(x) = \frac{1}{x-2} + 1$ is shown on a Cartesian coordinate system. The x-axis and y-axis are represented by thin grey lines. A horizontal dashed red line is drawn at $y = 1$, representing a horizontal asymptote. The function curve is a hyperbola with two branches. One branch is in the upper right region, starting from a vertical asymptote at $x = 2$ and approaching the horizontal asymptote $y = 1$ as x increases. The other branch is in the lower left region, also starting from the vertical asymptote at $x = 2$ and approaching the horizontal asymptote $y = 1$ as x decreases. The curve passes through the point $(3, 2)$ and $(1, 0)$.

$$f(x) = \frac{1}{x-2} + 1$$

$$y = 1$$